Do stricter gun laws reduce firearm deaths?

A 50-state comparison of 2024 law grades and mortality

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Stronger law grades coincide with lower mortality.
This comparison cannot establish a causal effect.

The measures and five-level law scale

Mortality

CDC age-adjusted firearm deaths per 100,000
12 months ending 2024 Q4
All firearm deaths, including suicides

Law strictness

GIFFORDS 2024 letter grades
1 F = most lax · 2 D = lax · 3 C = middle
4 B = strict · 5 A = strictest
+ and − variants stay in the source table

50 states

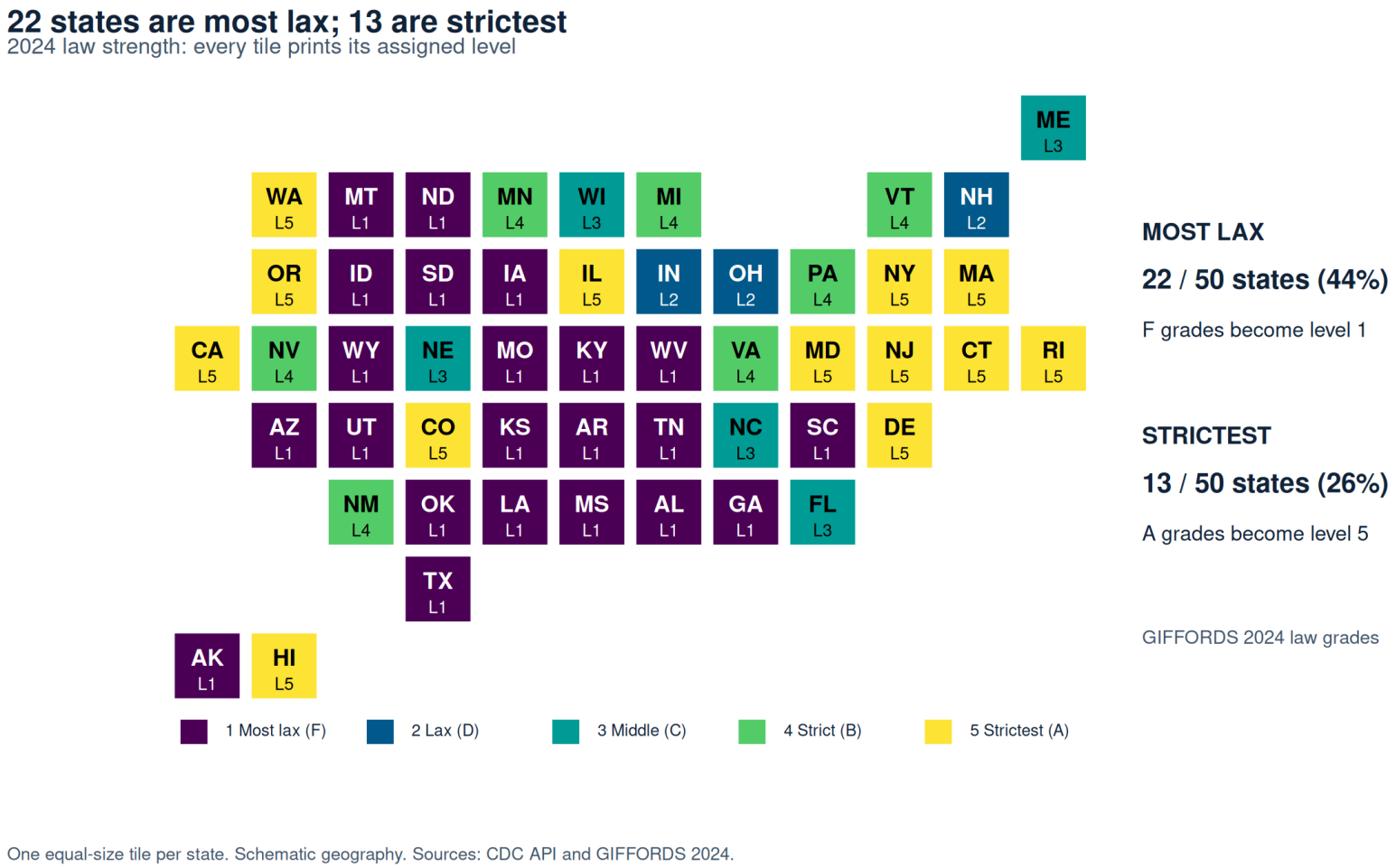
50 explicit law assignments and API rates
Group sizes: 22, 3, 5, 7, and 13 states

Why these choices

A common year and age-adjusted rates improve
comparability. The five bins preserve the source ordering without inventing state grades.

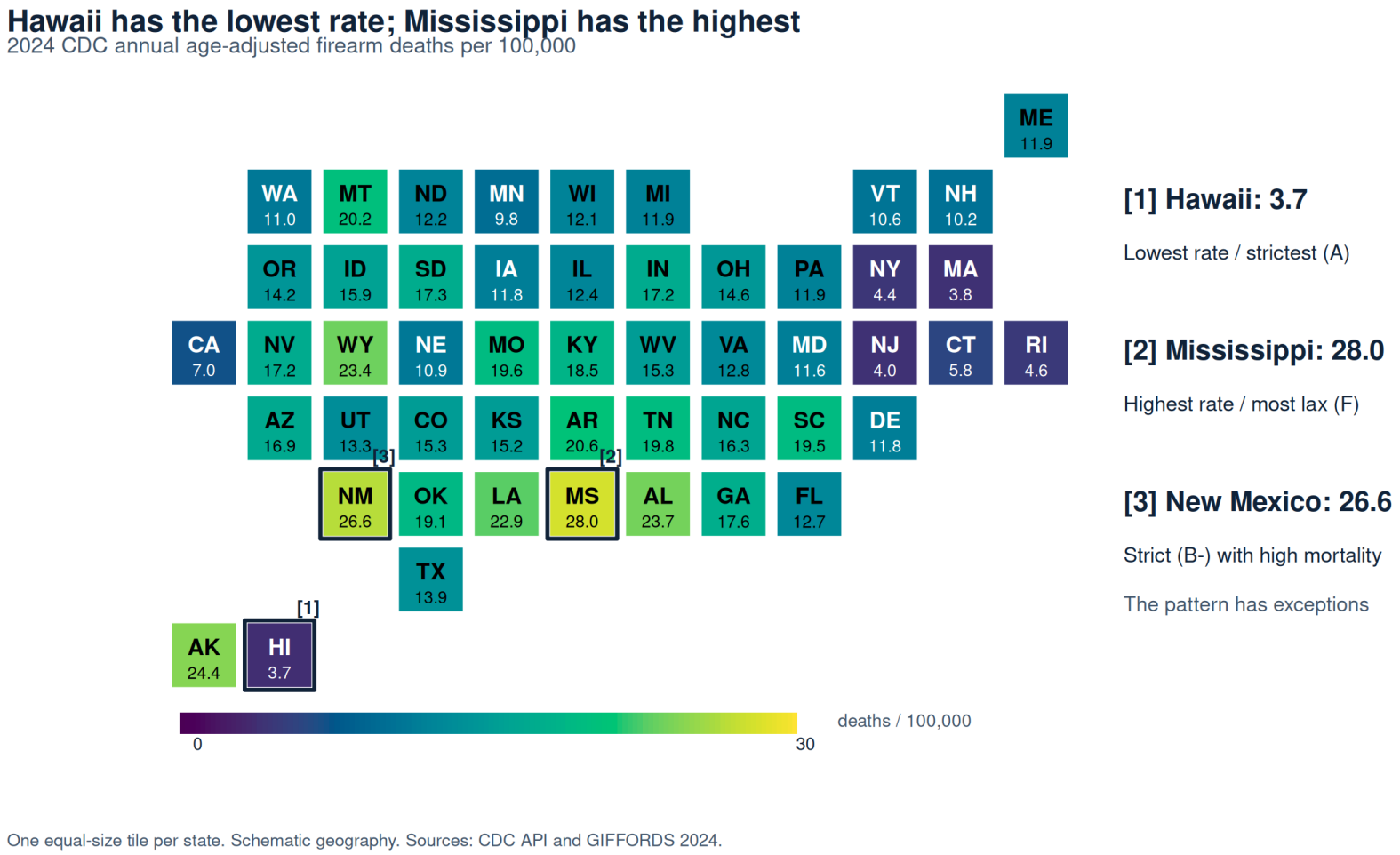
Sources: CDC NCHS open-data API (2024 Q4 annual age-adjusted rates); GIFFORDS 2024 scorecard.

22 states are most lax; 13 are strictest



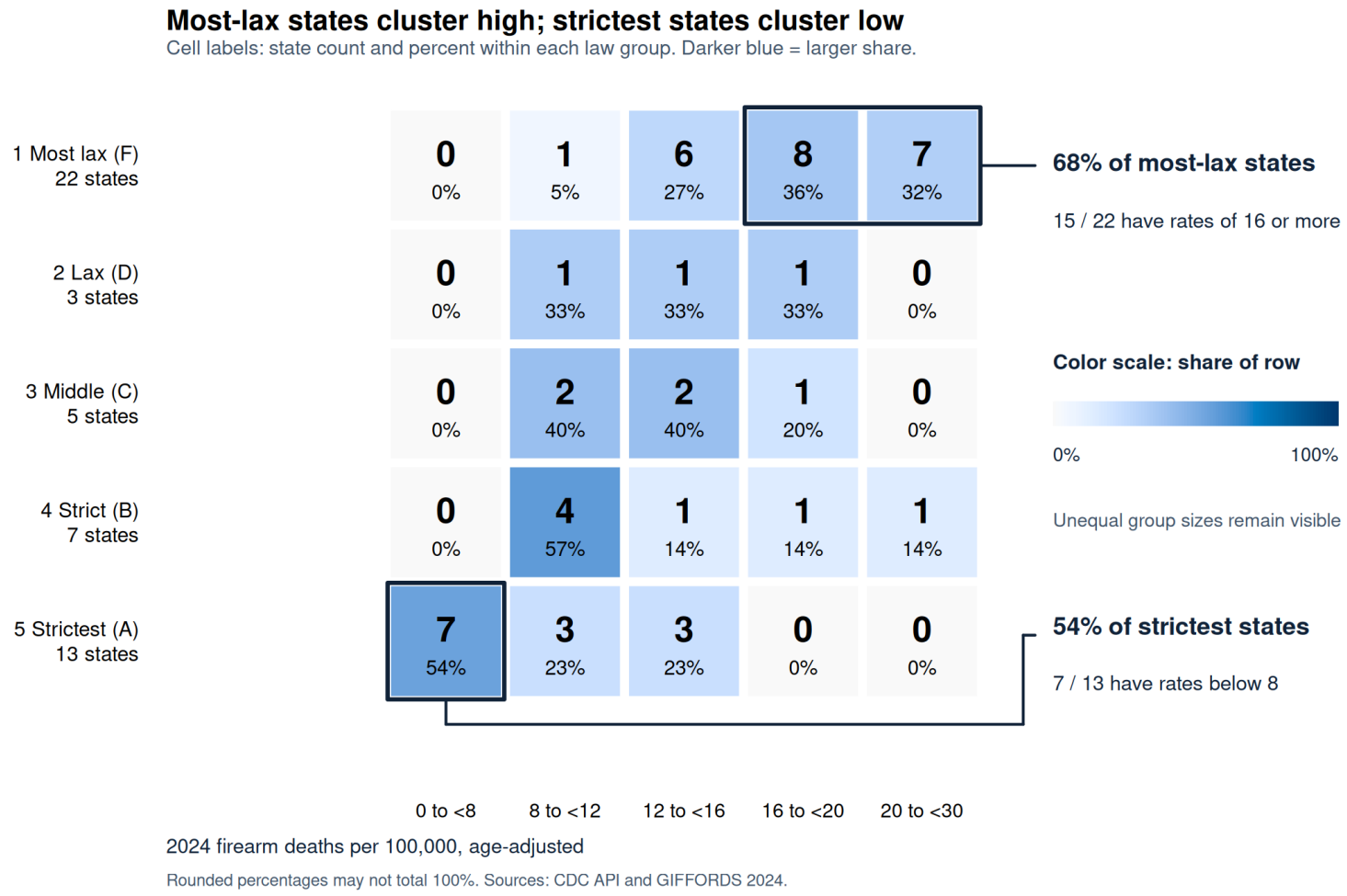
Equal tiles keep every state visible. Printed levels support the color legend.

Mortality extremes fit the pattern, with exceptions



Numbered outlines connect the extremes and exception directly to their state tiles.

Most lax states cluster in higher mortality bands



Within-group percentages allow comparison despite unequal group sizes. Counts retain the evidence.

The strictest group averages 8.4 deaths versus 18.6

Law level	States	Mean rate	Median rate
1 · Most lax (F)	**22**	**18.6**	**18.8**
2 · Lax (D)	3	14.0	14.6
3 · Middle (C)	5	12.8	12.1
4 · Strict (B)	7	14.4	11.9
5 · Strictest (A)	**13**	**8.4**	**7.0**

Spearman correlation: −0.717

Rates: 2024 age-adjusted deaths per 100,000. Each state has equal weight.
New Mexico raises the B-group mean.

The evidence supports an association

Observed answer

Stronger 2024 law grades tend to accompany lower 2024 age-adjusted firearm mortality.

Limits that affect the conclusion

Other state characteristics and policy timing remain uncontrolled.
The law grades come from one advocacy organization.
All firearm deaths combine several intents with different causes.

A stronger causal test would track law changes and mortality across years with suitable comparison states and controls.

Sources, visual choices and reproducibility

CDC NCHS API

data.cdc.gov/resource/489q-934x.json
Annual 2024 age-adjusted firearm-related injury rates

GIFFORDS 2024 scorecard

giffords.org/resources/scorecard2024/
Law grades recoded to five ordered bins; original grades retained

R heat maps and complete supporting files

Labeled equal-size tiles and accessible palettes make the states comparable.
The package includes API evidence, source code, all 50 state assignments,
the notebook, PNG/PDF figures and written explanations.

AI assisted with coding, analysis and editorial review.
The accompanying disclosure documents this assistance.